Semantic typology of color naming

Computational modeling of language
Spring 2026

Bureaucracy

- Next week will be our **midterm exam**:
  - Tuesday: review session
  - Thursday: exam itself

Thank you for the feedback last time!

Good

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- “I’m learning a lot.”

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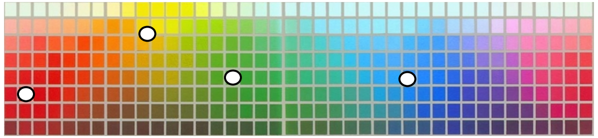
- Thursdays for more lecture, instead of homeworks and related “doing”?
- Theory-heavy lectures and application-oriented homeworks feel separate.
- What are course goals, and how does our day-to-day activity advance those goals?
- How to study for the midterm exam?

Color categories across languages

Universal foci

Categories organized around 6 **universal foci**: black, white, red, green, yellow, blue. (Berlin & Kay, 1969; Heider, 1972).



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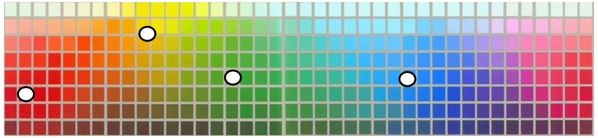

Foci universally **privileged** in both **naming** and **memory**.

Universal foci

Naming: Implicational hierarchy (Berlin & Kay, 1969)

$$\begin{bmatrix}\text{white}\\ \text{black}\end{bmatrix} < [\text{red}] < \begin{bmatrix}\text{green}\\ \text{yellow}\end{bmatrix} < [\text{blue}] < [\text{brown}] < \begin{bmatrix}\text{purple}\\ \text{pink}\\ \text{orange}\\ \text{grey}\end{bmatrix}$$

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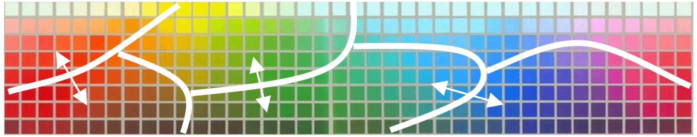
Memory: Focal colors remembered better than others, even by speakers of languages with dissimilar color naming systems (Heider, 1972).

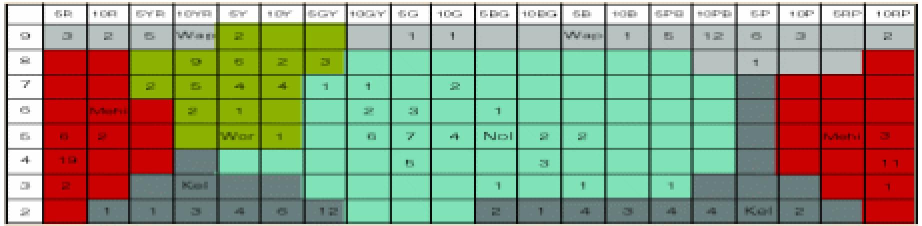
Relativist

No universal foci: categories organized at **boundaries** by linguistic convention. (Roberson, Davies, & Davidoff, 2000; Roberson, Davidoff, Davies, & Shapiro, 2005).

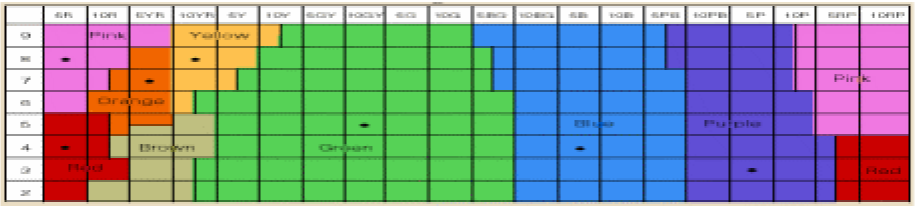


<u>Relativist</u>

Berinmo



English



Failure to replicate better memory at foci.

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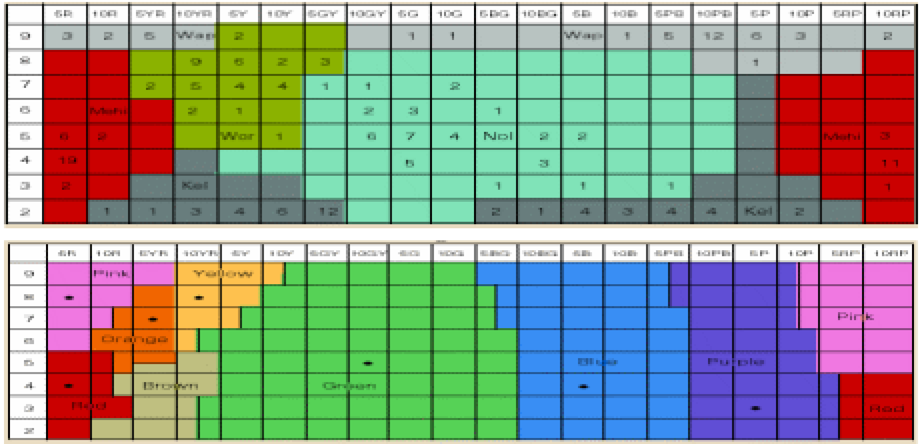

Failure to replicate better memory at foci.
Instead, better discrimination from memory at **boundaries**...
…which **vary** across languages (Roberson et al. 2000).

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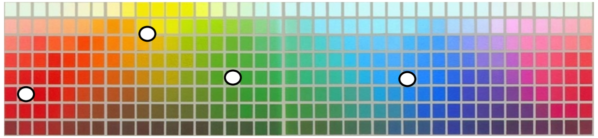

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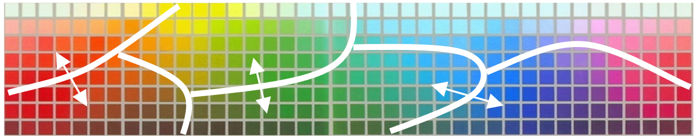

Boundaries free to **vary**, and **memory varies** with them.

Two **separate** but **related** questions

1. Does color cognition reflect one's native language? (Sapir-Whorf hypothesis)

2. Does color naming vary arbitrarily across languages? (Semantic typology)

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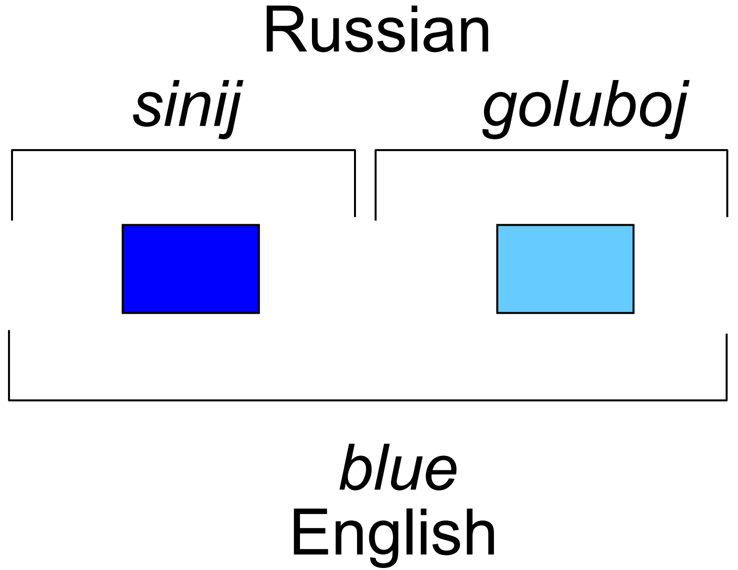

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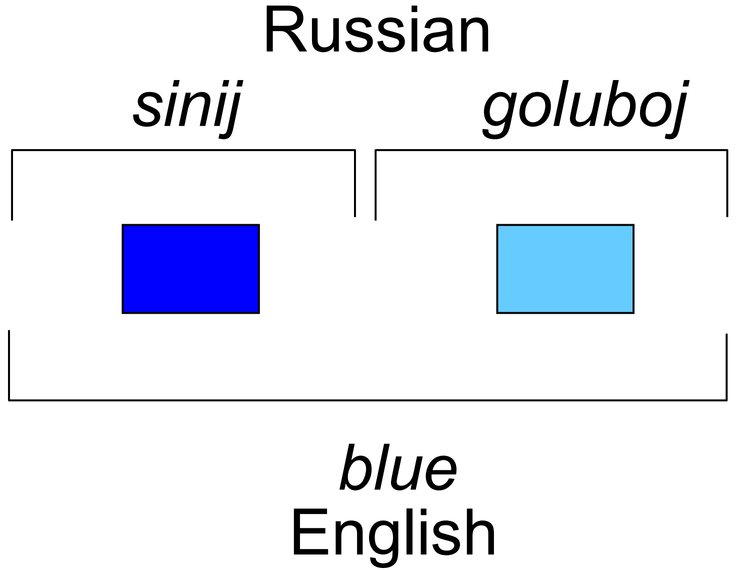
Enhanced color discrimination at language-defined boundaries.



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Tarahumara (Kay & Kempton, 1984)
Berinmo (Roberson et al., 2000)
Himba (Roberson et al., 2005)
Russian (Winawer et al., 2007)
Korean (Roberson et al., 2008)
…

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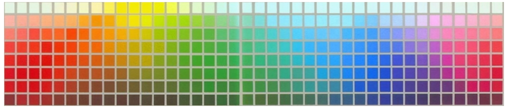
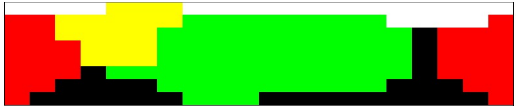
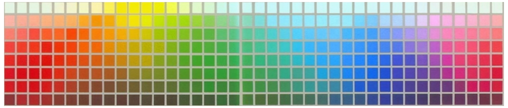
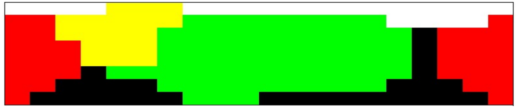




Berinmo: Papua New Guinea





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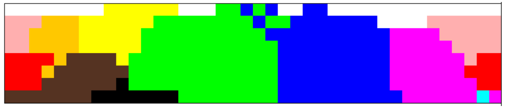
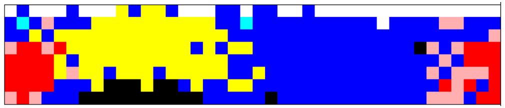
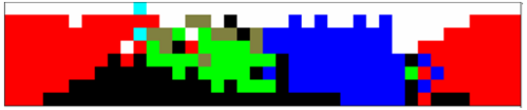


Chavacano: Philippines

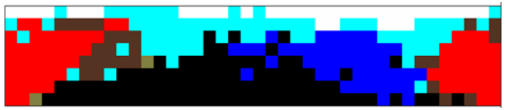
Wobé: Ivory Coast



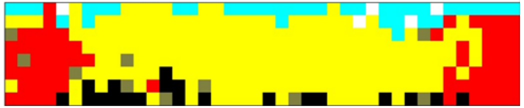
Eastern Cree: Canada



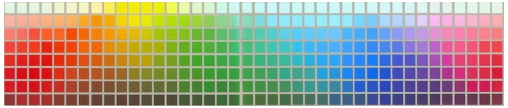
Gunu: Cameroon

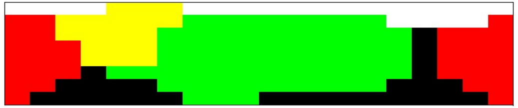


Abidji: Ivory Coast



Karajá: Brazil





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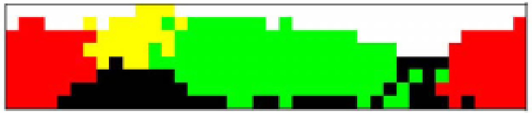
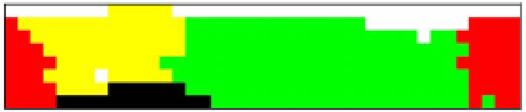
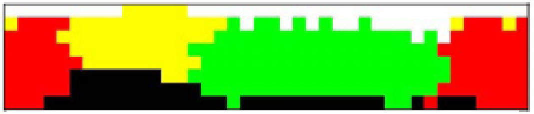


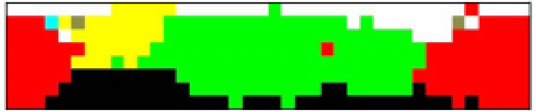
Bauzi: Irian Jaya



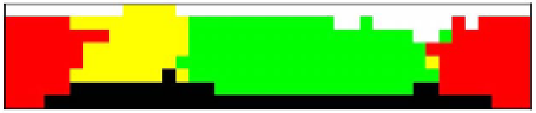
Paya: Honduras



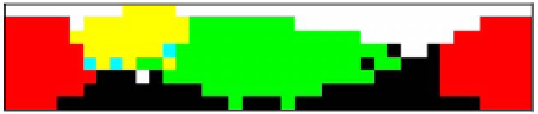
Sirionó: Bolivia



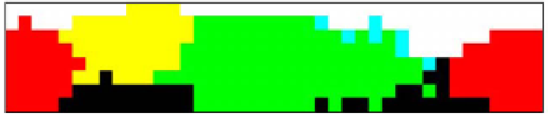
Iwam: Papua New Guinea



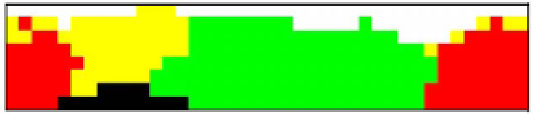
Yaminahua: Peru



Berik: Irian Jaya



Colorado: Ecuador



Jicaque: Honduras

Focal colors are universal after all

Terry Regier[†‡§], Paul Kay[‡¶], and Richard S. Cook[¶]

Proceedings of the National Academy of Sciences, 2005

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- But these studies had primarily investigated languages of **industrialized** societies.
- Roberson et al. argued **against** universality, using data from a **non-industrialized** society, **Berinmo**.
- We examined a **broad range** of languages of **non-industrialized** societies.

Table 1. WCS languages, families, and countries where encountered

Index	Language	Family	Country where encountered
1	Abidji	Kwa	Ivory Coast
2	Agarabi	Trans-New Guinea	Papua New Guinea
3	Agta	Austronesian	Philippines
4	Aguacatec	Mayan	Guatemala
5	Amarakaeri	Arawakan	Peru
6	Ampeeli	Angan	Papua New Guinea
7	Amuzgo	Oto-Manguean	Mexico
8	Angaatiha	Angan	Papua New Guinea
9	Apinayé	Macro-Ge	Brazil
10	Arabela	Zaparoan	Peru
11	Bahinemo	Sepik Hill	Papua New Guinea
12	Bauzi	Geelvink Bay	Indonesia
13	Berik	Trans-New Guinea	Indonesia (Irian Jaya)
14	Bété	Kru	Ivory Coast
15	Bhili	Indic	India
	...		
105	Yacouba	Dan	Ivory Coast
106	Yakan	Austronesian	Philippines
107	Yaminahua	Panoan	Peru
108	Yucuna	Arawakan	Colombia
109	Yupik	Eskimo-Aleut	United States
110	Zapotec	Oto-Manguean	Mexico

The World Color Survey

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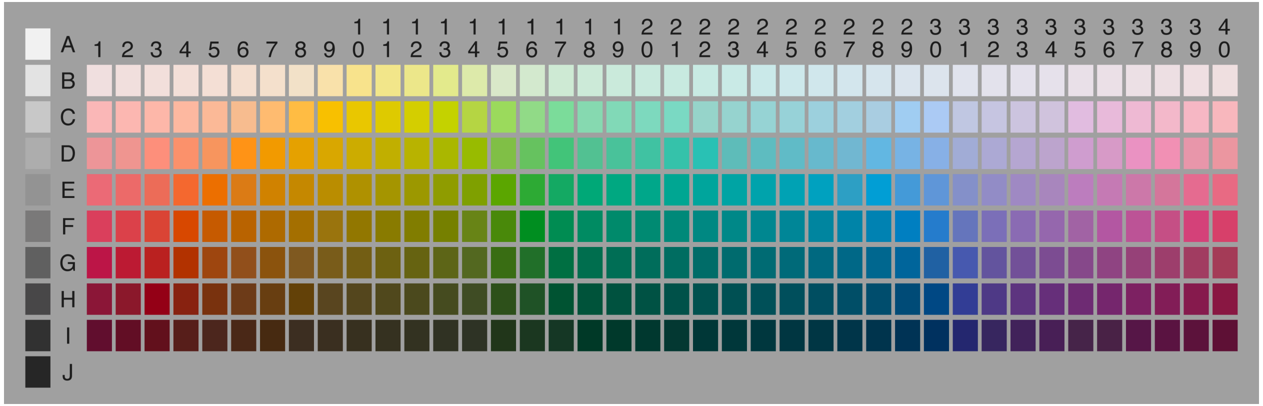
- Collected in the 1970s; web-posted in the early 2000s; now a widely-used data resource.

The World Color Survey



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- **Naming data**: What would you call **this** color?

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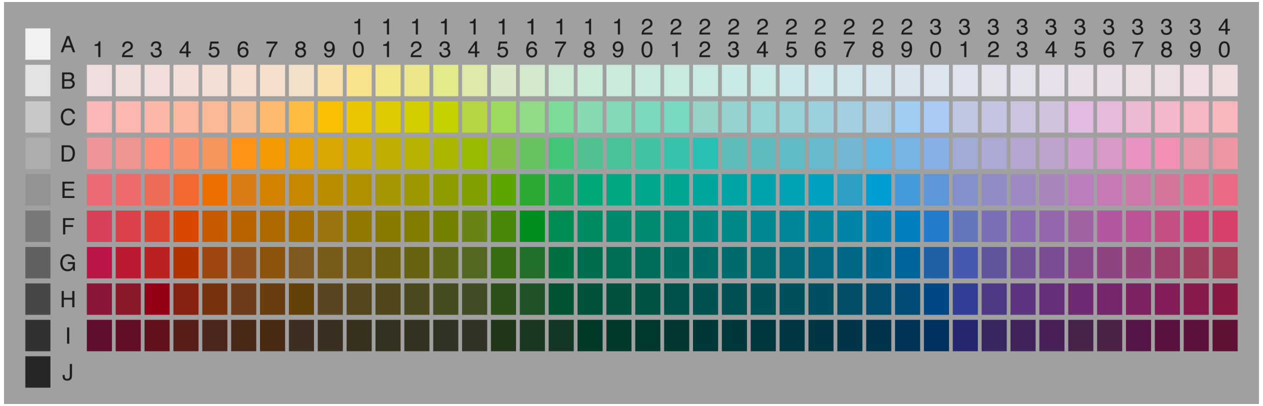

- **Naming data**: What would you call **this** color?
- **Focus data**: Please show me the **best example** of <color term>.

Hypotheses

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- **Universalist**: best examples of color terms reflect the 6 proposed universal foci: black, white, red, green, yellow, and blue.

Hypotheses

- **Universalist**: best examples of color terms reflect the 6 proposed universal foci: black, white, red, green, yellow, and blue.
- **Relativist**: "color categories are developed from demarcation at boundaries…the central tendency of exemplars can be extracted at a later stage" (Roberson et al., 2000).

Prediction

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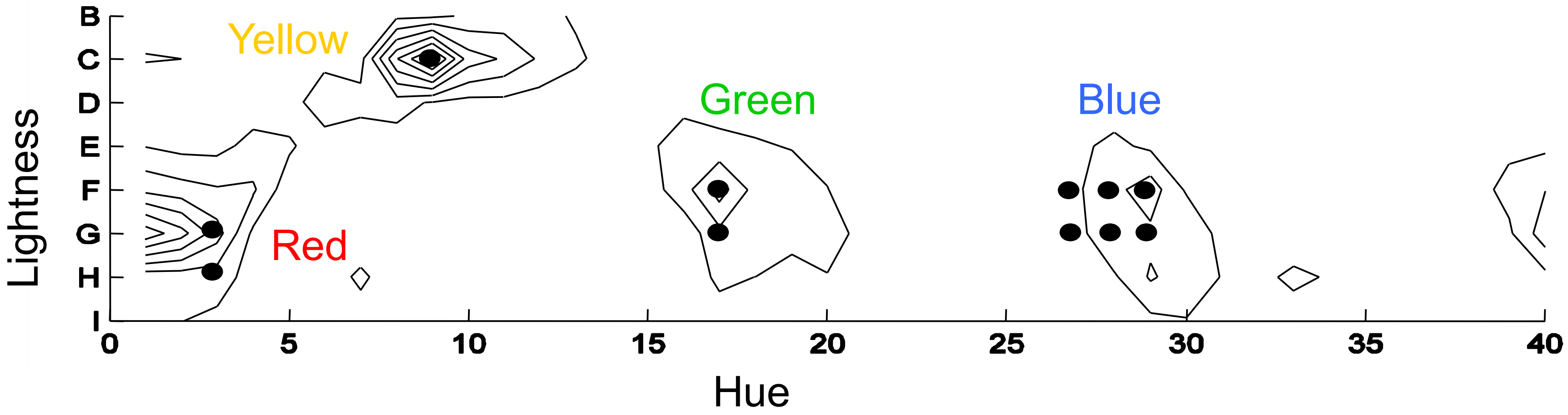
- **Relativist**: we should not expect to see clustering of this specific sort.

“We pooled all of the focus data from all speakers in all languages of the WCS and calculated how many best-example choices (hits) fell on each chip of the array. The two chips in the array that received the most hits were A0 (2,048 hits) [best example of English *white*] and J0 (1,988 hits) [best example of English *black*].”



Best examples of color terms from 110 languages of non-industrialized societies – compared with English.

Regier, Kay, & Cook (2005). *PNAS*.

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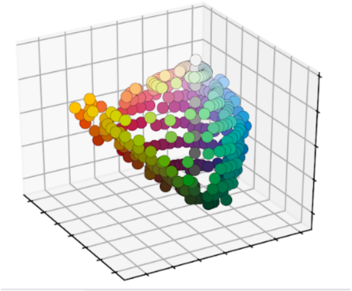
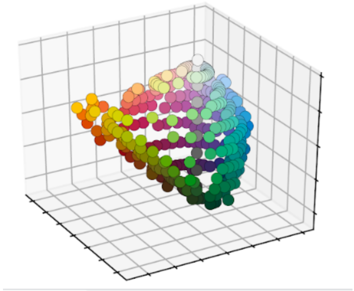
- **Universalist**: best examples should cluster more tightly across languages than do the centers of category extensions, because category extension is known to vary across languages.
- **Relativist**: best examples should not cluster more tightly than the centers of category extensions, because on this view, best examples <u>are</u> category centers.



CIELAB: a 3-dimensional color space in which Euclidean distance approximates **perceptual dissimilarity**.



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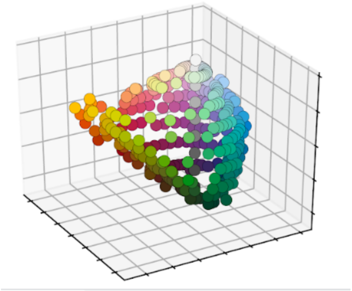
- *c(t)*: **Centroid** of color term *t*, in CIELAB



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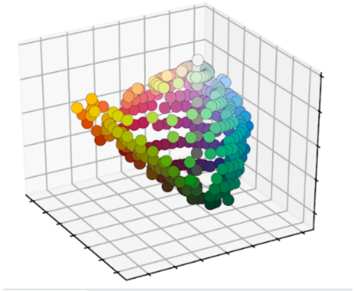

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Compared across **non-industrialized** (WCS) and largely **industrialized** (Berlin & Kay) languages.

$$CS_l = \sum_{t \in l} \sum_{l^* \in BK} \min_{t^* \in l^*} \text{distance}[c(t), c(t^*)]$$

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<u>CS_l : centroid separation</u>

For each term *t* in WCS language *l*, find the closest term *t** in each B&K language *l**, and sum the distances.

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<u>*CS*</u>

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$$FS_l = \sum_{t \in l} \sum_{l^* \in BK} \min_{t^* \in l^*} \text{distance}[f(t), f(t^*)]$$

FS_l: focus separation : the same, only with foci.

A paired *t* test revealed that the focus separation [*FS*] (M = 5,596.98) was smaller than the centroid separation [*CS*] (*M* = 6,391.78) [$t(109) = 10.2506$; $P<0.0001$].

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Foci align better than centroids across languages.

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Two **separate** but **related** questions

1. Does color cognition reflect one's native language? (Sapir-Whorf hypothesis)

2. Does color naming vary arbitrarily across languages? (Semantic typology)

What **questions** do you have?

What **comments** do you have?

- End of the story? **No**!

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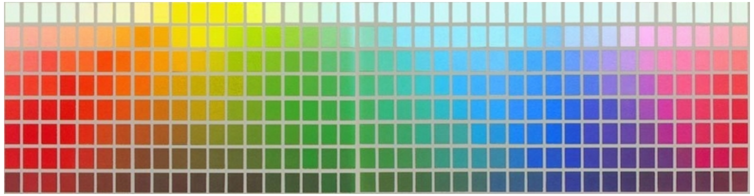
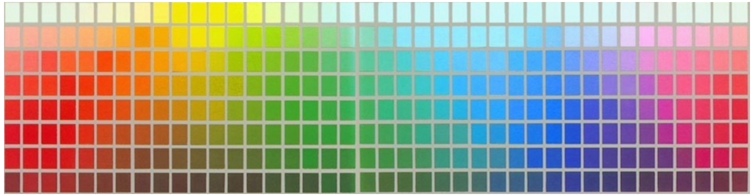
- End of the story? **No**!
- There is **variation**, both in boundaries and in foci.
- The variation is **constrained**, but real.

Color categories across languages

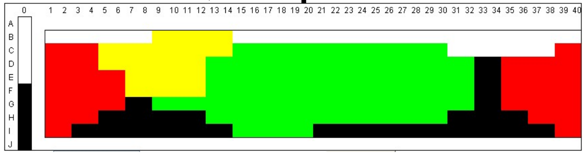


Color categories across languages

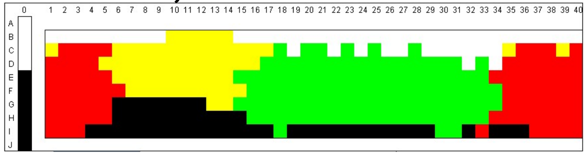


Universals

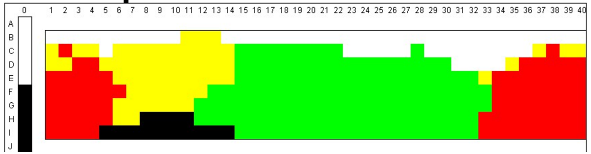
Berinmo, Papua New Guinea



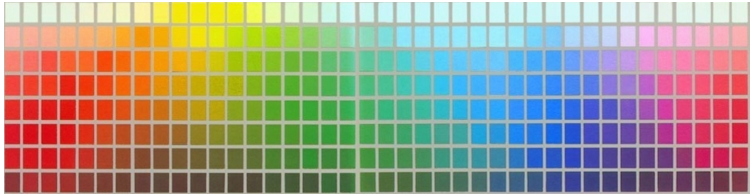
Sirionó, Bolivia



Jicaque, Honduras



Color categories across languages



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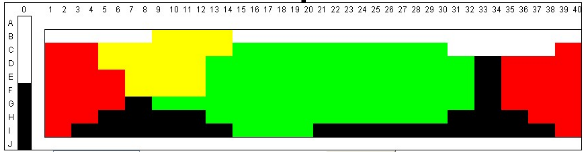
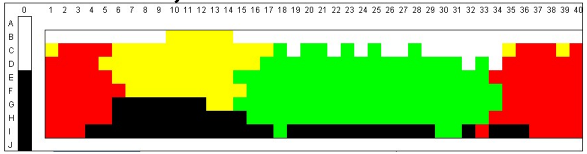
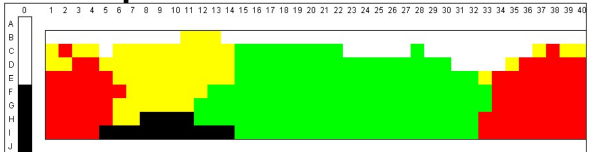


Sirionó, Bolivia

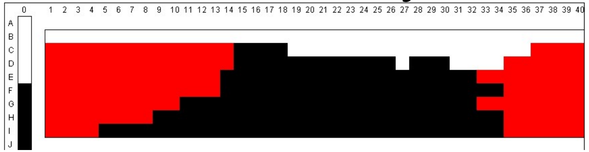


Jicaque, Honduras

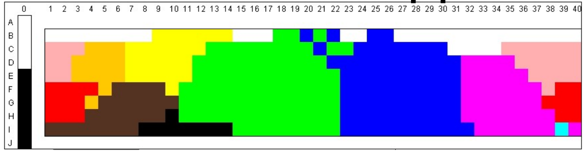


Variation

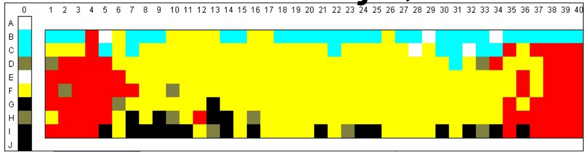
Wobé, Ivory Coast



Chavacano, Philippines



Karajá, Brazil



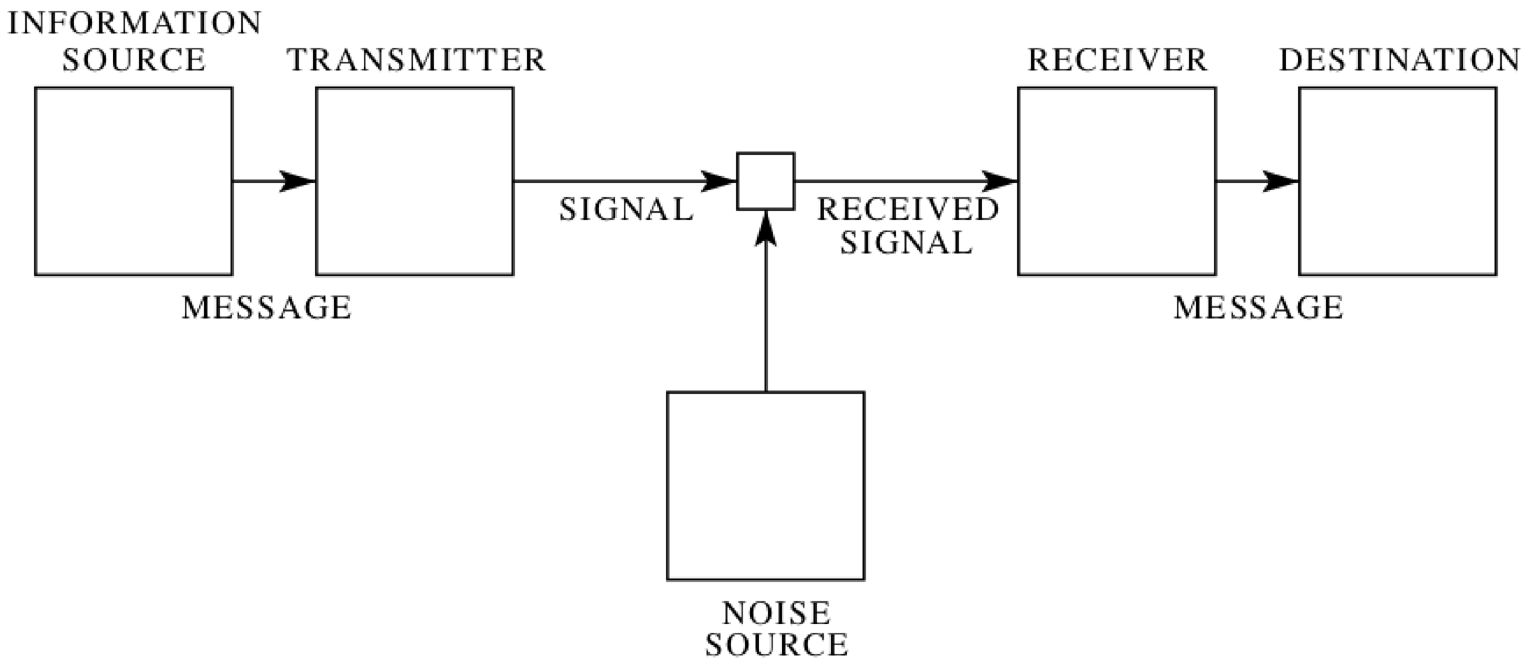
How can we account for this pattern of **constrained variation**?

Efficient compression in color naming and its evolution

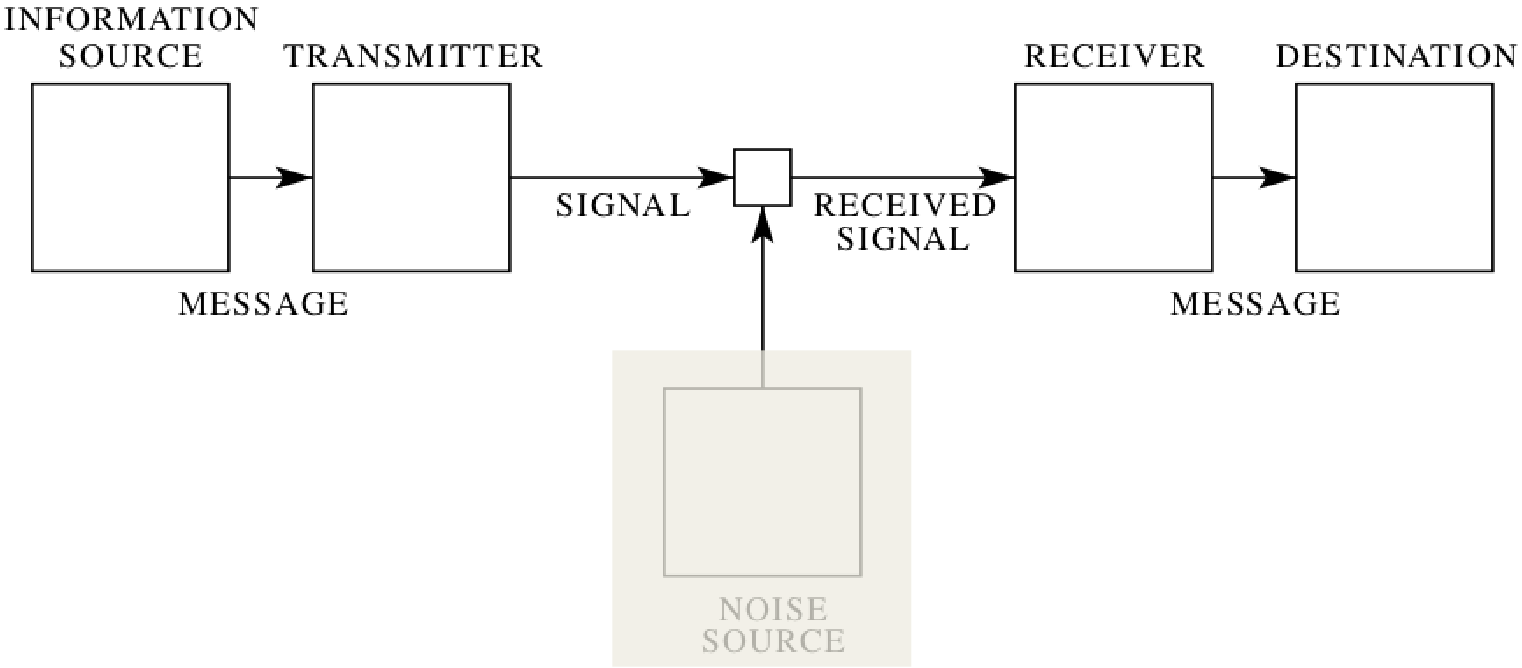
Noga Zaslavsky[a,b,1], Charles Kemp[c,2], Terry Regier[b,d], and Naftali Tishby[a,e]

Proceedings of the National Academy of Sciences, 2018

Noga Zaslavsky

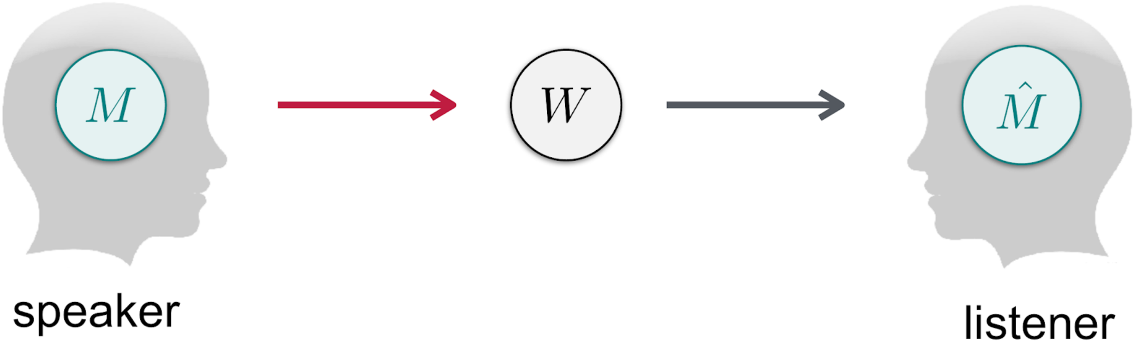


Shannon, 1948

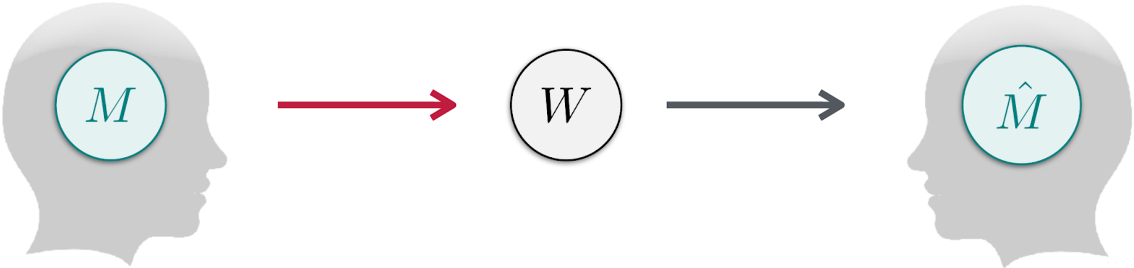


Shannon, 1948

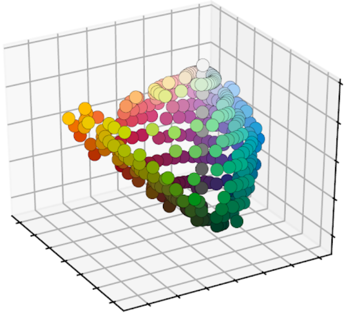
Communication model



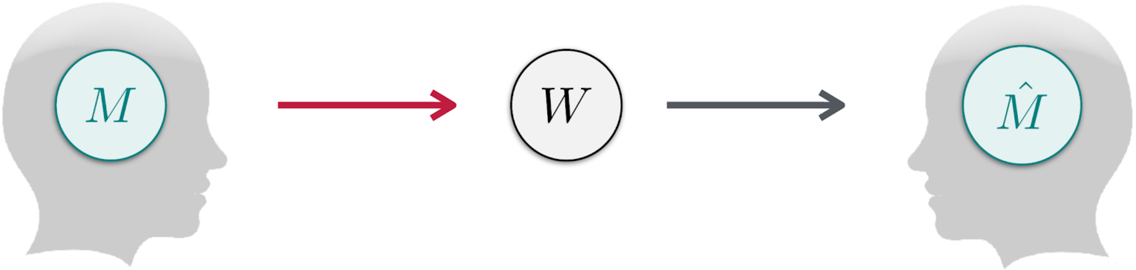
Communication model



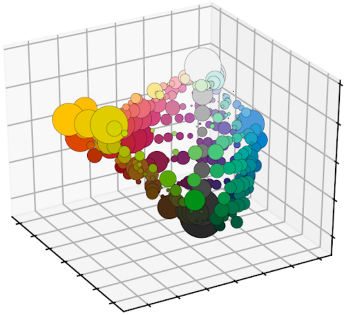
- Perceptually grounded meanings
  [Regier et al., 2007, 2015]



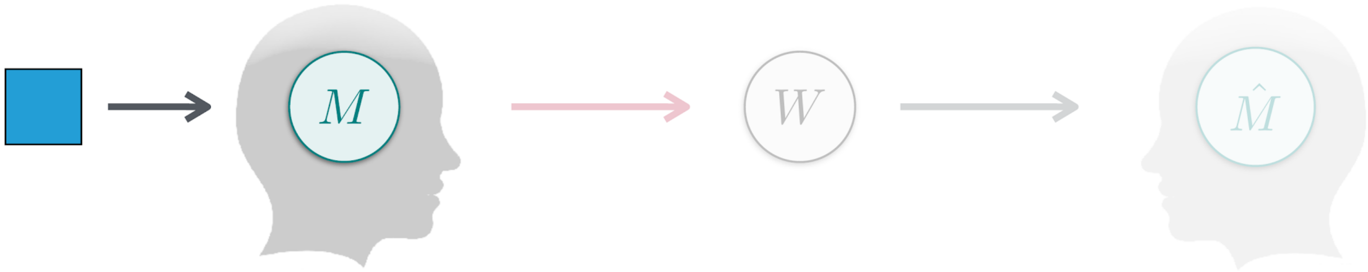
Communication model



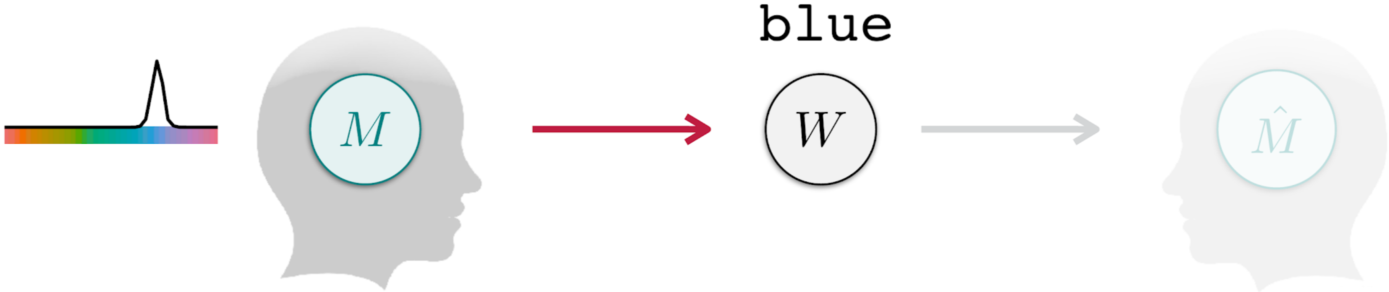
- Perceptually grounded meanings
  [Regier et al., 2007, 2015]
- Communicative need
  [Zaslavsky et al., 2018]



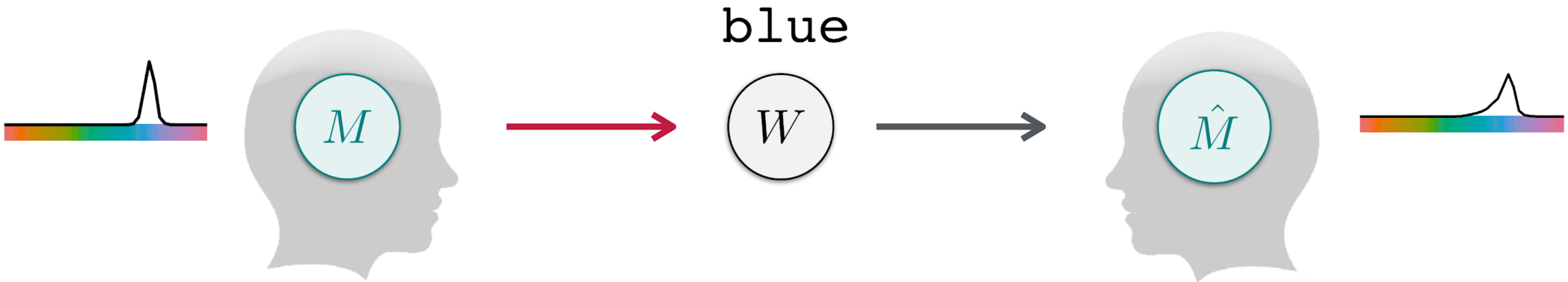
Communication model



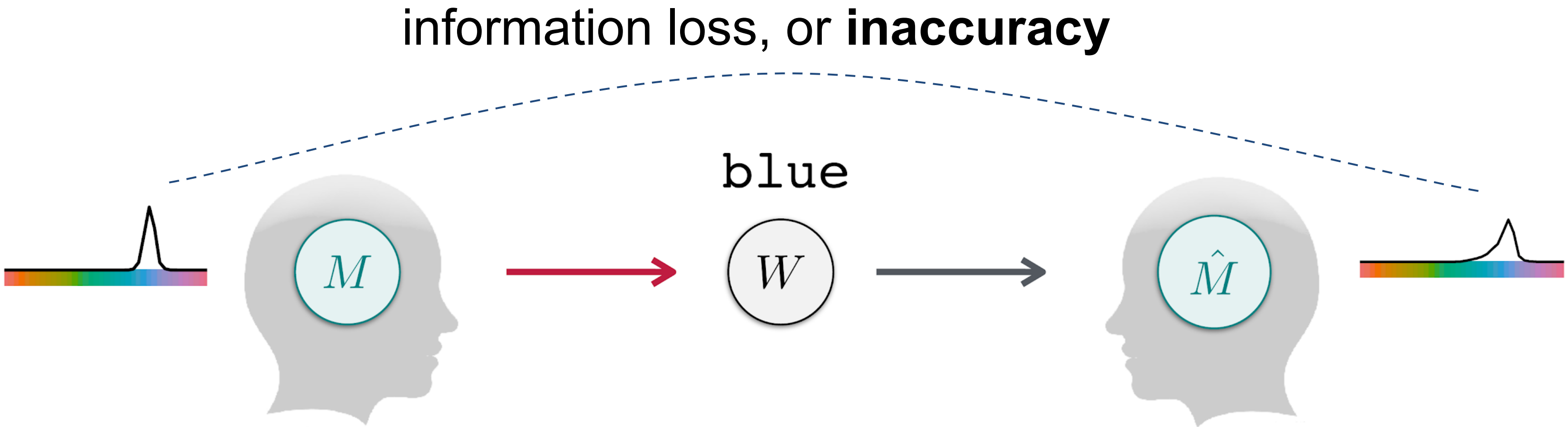
Communication model



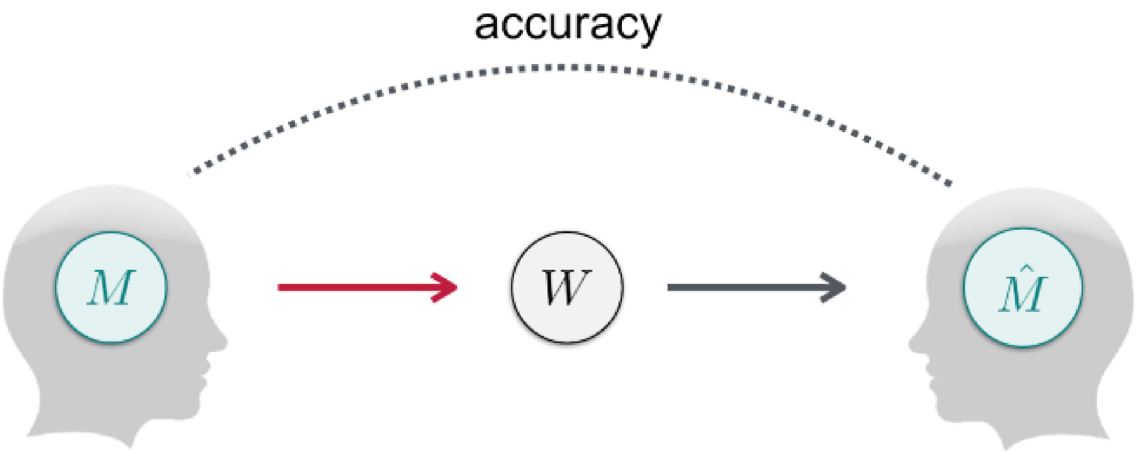
Communication model



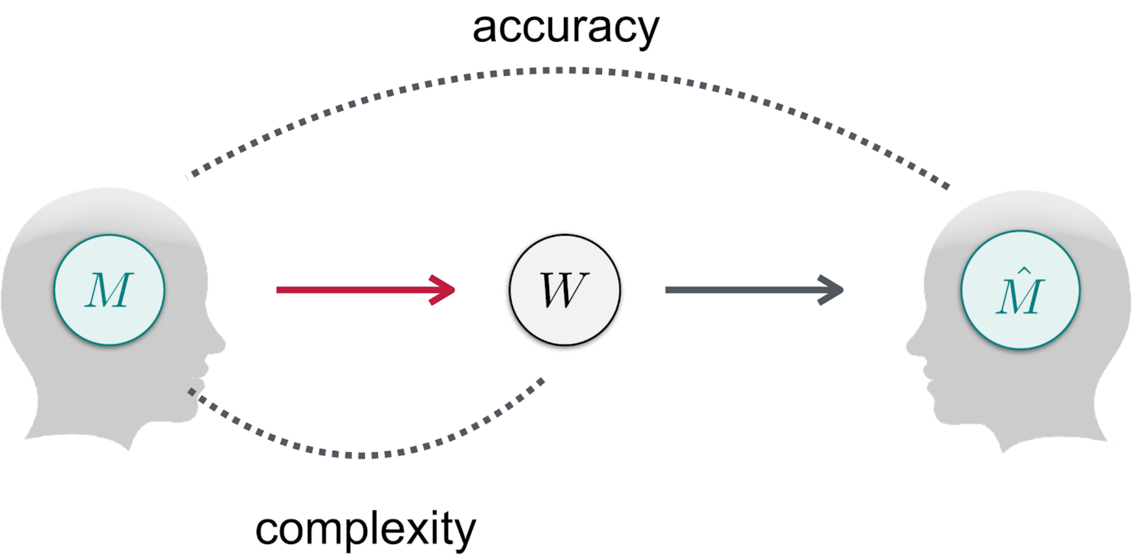
Communication model



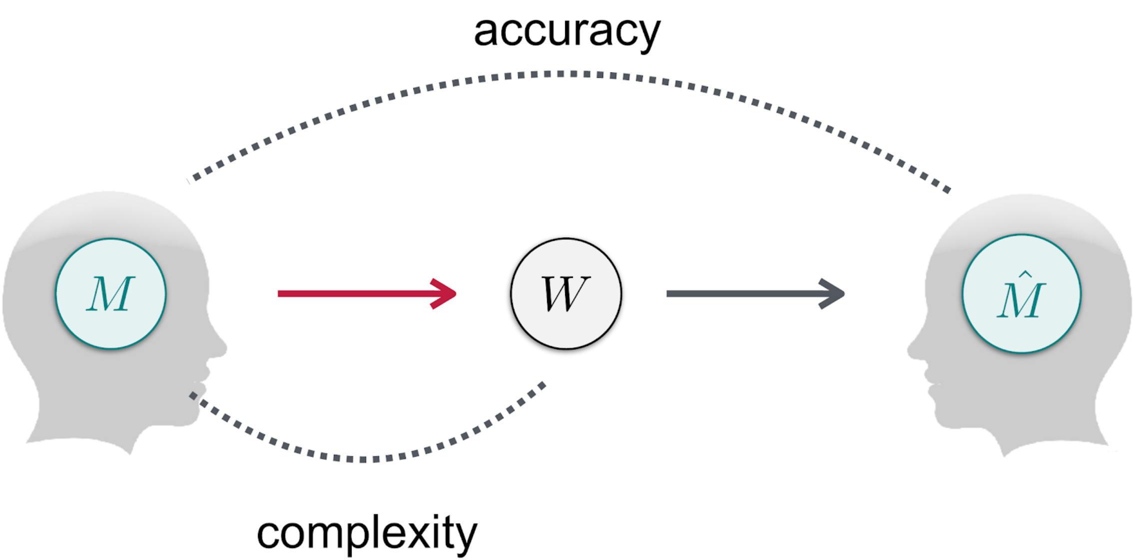
Efficient communication



Efficient communication



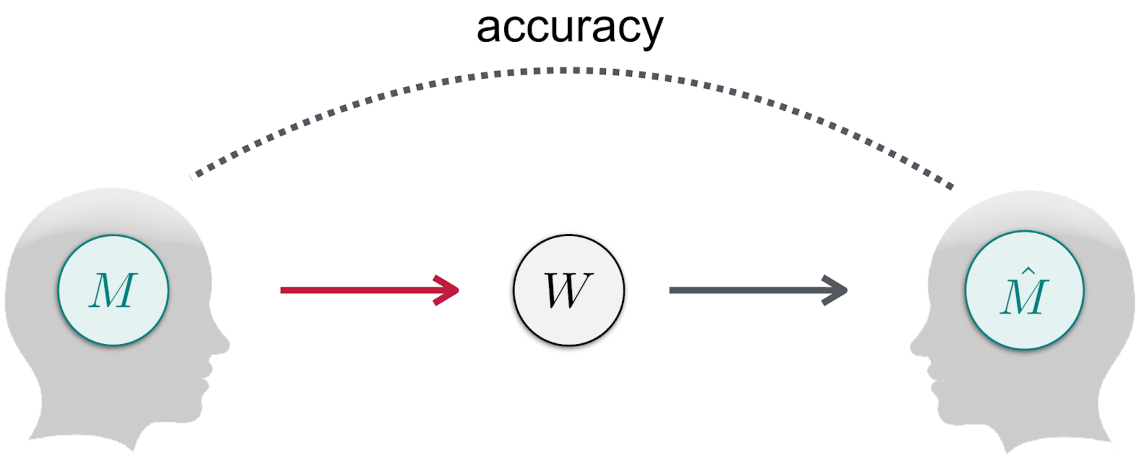
The Information Bottleneck (IB) principle



Tishby et al., 1999

Zaslavsky et al., *PNAS*, 2018

The Information Bottleneck (IB) principle

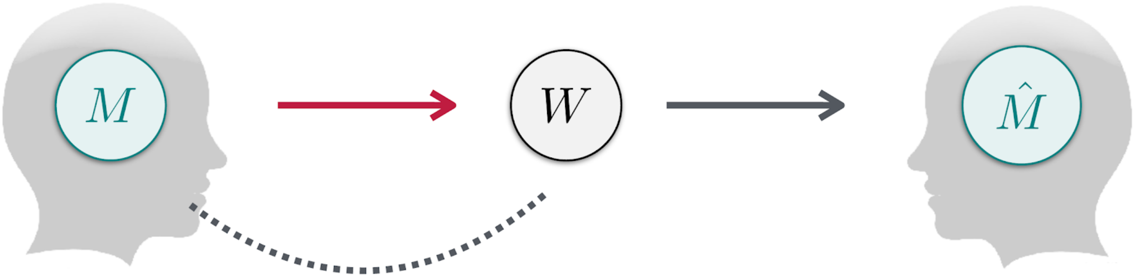


High accuracy means low distortion, $\mathbb{E}[D[M\|\hat{M}]]$

Tishby et al., 1999

Zaslavsky et al., *PNAS*, 2018

The Information Bottleneck (IB) principle

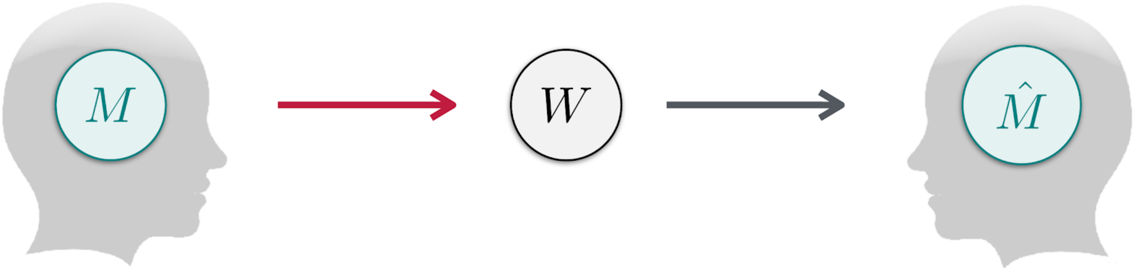


complexity $= I(M;W)$

Tishby et al., 1999

Zaslavsky et al., *PNAS*, 2018

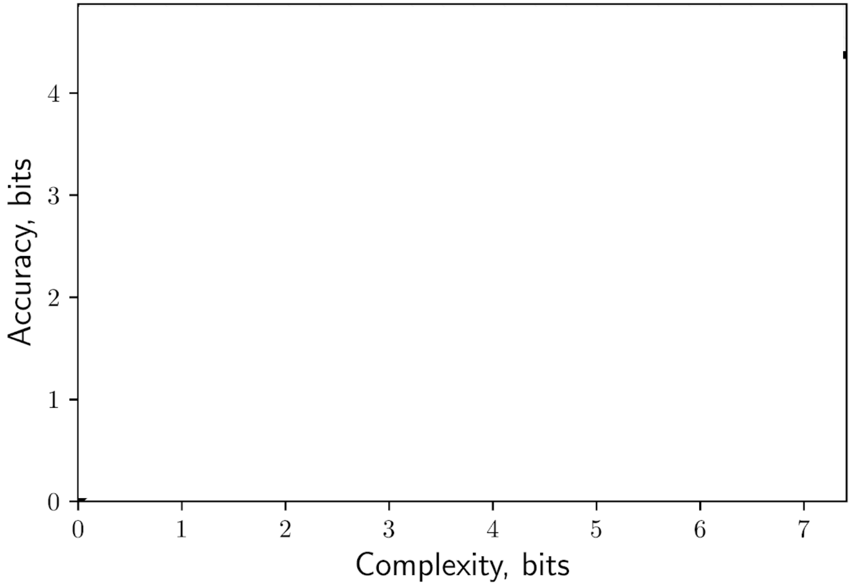
The Information Bottleneck (IB) principle



Optimal **systems** minimize: **complexity** $-\beta$ **accuracy**

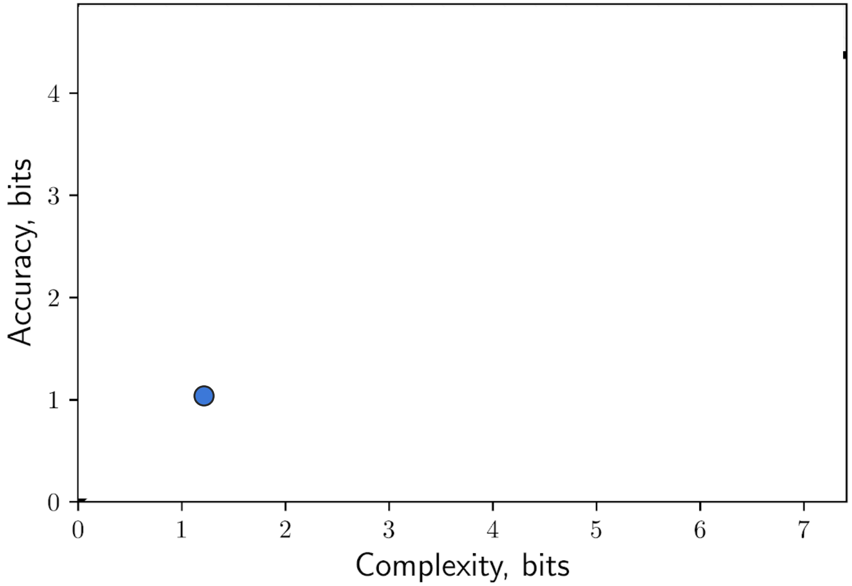
tradeoff

The information plane

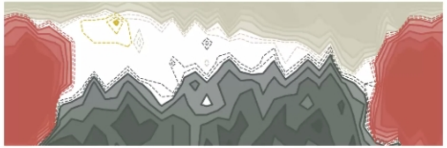


Zaslavsky et al., *PNAS*, 2018

The information plane

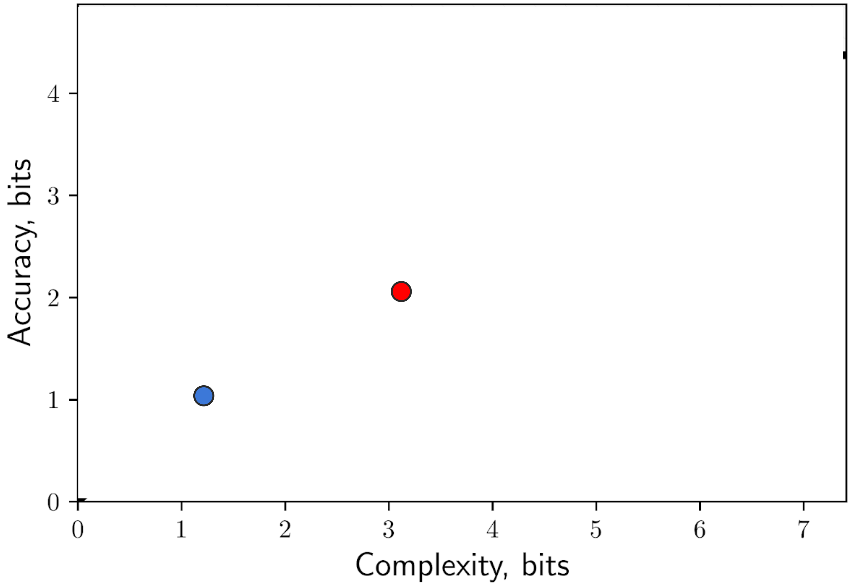


Kwerba
(iso: kwe, Indonesia)



Zaslavsky et al., *PNAS*, 2018

The information plane



Artificial, hue-based

Kwerba
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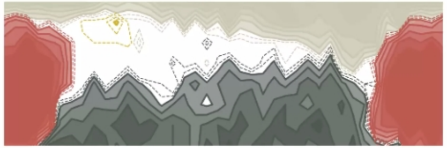
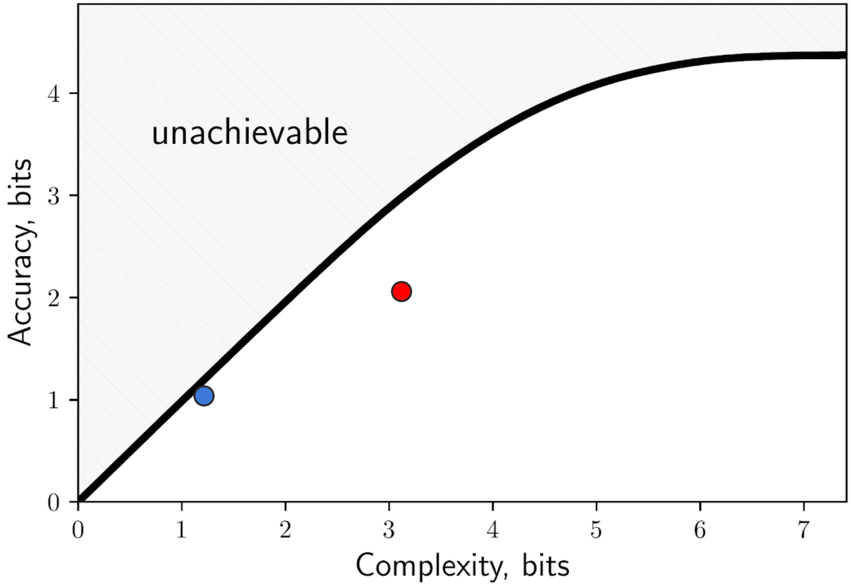


Zaslavsky et al., *PNAS*, 2018

The information plane



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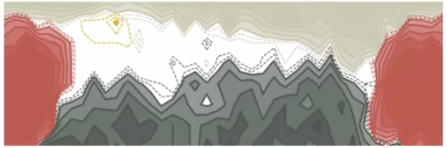
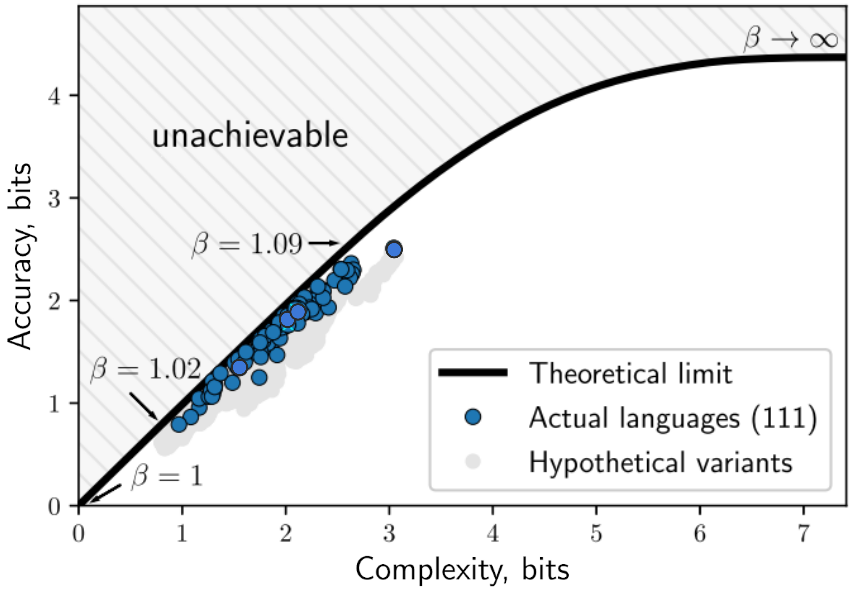


Zaslavsky et al., *PNAS*, 2018

Languages are near-optimally efficient



110 languages of the World Color Survey, + English

Zaslavsky et al., *PNAS*, 2018

Explaining known descriptive generalizations

$$\begin{bmatrix}\text{white}\\ \text{black}\end{bmatrix} < [\text{red}] < \begin{bmatrix}\text{green}\\ \text{yellow}\end{bmatrix} < [\text{blue}] < [\text{brown}] < \begin{bmatrix}\text{purple}\\ \text{pink}\\ \text{orange}\\ \text{grey}\end{bmatrix}$$

Berlin & Kay 1969

Explaining known descriptive generalizations

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Berlin & Kay 1969

while addressing a known critique

Optimal:  Levinson 2001

Explaining known descriptive generalizations

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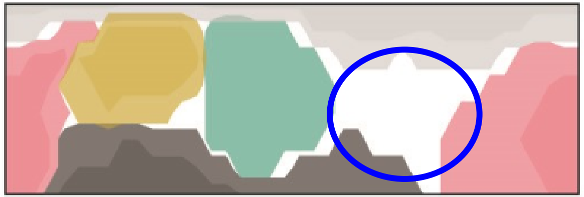
Berlin & Kay 1969

while addressing a known critique

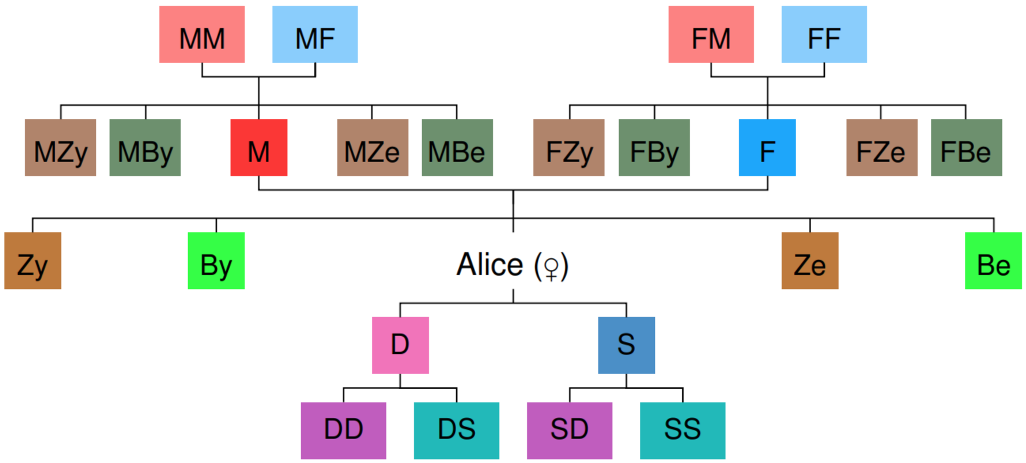
Optimal:  Levinson 2001

- Universals **and** variation in color naming across languages are explained by the principle of **efficient communication**.

- Universals **and** variation in color naming across languages are explained by the principle of **efficient communication**.

- A **single parameter** explains much of the variation.

Kinship



$\texttt{mother}(x, y) \leftrightarrow \texttt{PARENT}(x, y) \wedge \texttt{FEMALE}(x)$

$\texttt{father}(x, y) \leftrightarrow \texttt{PARENT}(x, y) \wedge \texttt{MALE}(x)$

$\texttt{daughter}(x, y) \leftrightarrow \texttt{CHILD}(x, y) \wedge \texttt{FEMALE}(x)$

$\texttt{son}(x, y) \leftrightarrow \texttt{CHILD}(x, y) \wedge \texttt{MALE}(x)$

$\texttt{sister}(x, y) \leftrightarrow \exists z\ \texttt{daughter}(x, z) \wedge \texttt{PARENT}(z, y)$

$\texttt{brother}(x, y) \leftrightarrow \exists z\ \texttt{son}(x, z) \wedge \texttt{PARENT}(z, y)$

$\texttt{sibling}(x, y) \leftrightarrow \exists z\ \texttt{CHILD}(x, z) \wedge \texttt{PARENT}(z, y)$

$\texttt{aunt}(x, y) \leftrightarrow \exists z\ \texttt{sister}(x, z) \wedge \texttt{PARENT}(z, y)$

$\texttt{uncle}(x, y) \leftrightarrow \exists z\ \texttt{brother}(x, z) \wedge \texttt{PARENT}(z, y)$

$\texttt{niece}(x, y) \leftrightarrow \exists z\ \texttt{daughter}(x, z) \wedge \texttt{sibling}(z, y)$

$\texttt{nephew}(x, y) \leftrightarrow \exists z\ \texttt{son}(x, z) \wedge \texttt{sibling}(z, y)$

$\texttt{grandmother}(x, y) \leftrightarrow \exists z\ \texttt{mother}(x, z) \wedge \texttt{PARENT}(z, y)$

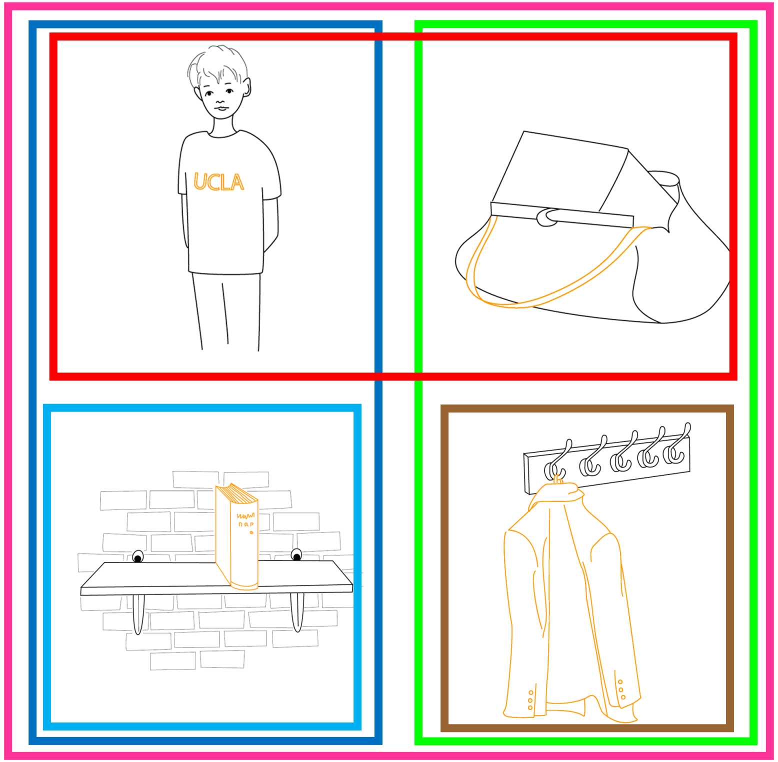
$\texttt{grandfather}(x, y) \leftrightarrow \exists z\ \texttt{father}(x, z) \wedge \texttt{PARENT}(z, y)$

$\texttt{granddaughter}(x, y) \leftrightarrow \exists z\ \texttt{daughter}(x, z) \wedge \texttt{CHILD}(z, y)$

$\texttt{grandson}(x, y) \leftrightarrow \exists z\ \texttt{son}(x, z) \wedge \texttt{CHILD}(z, y)$

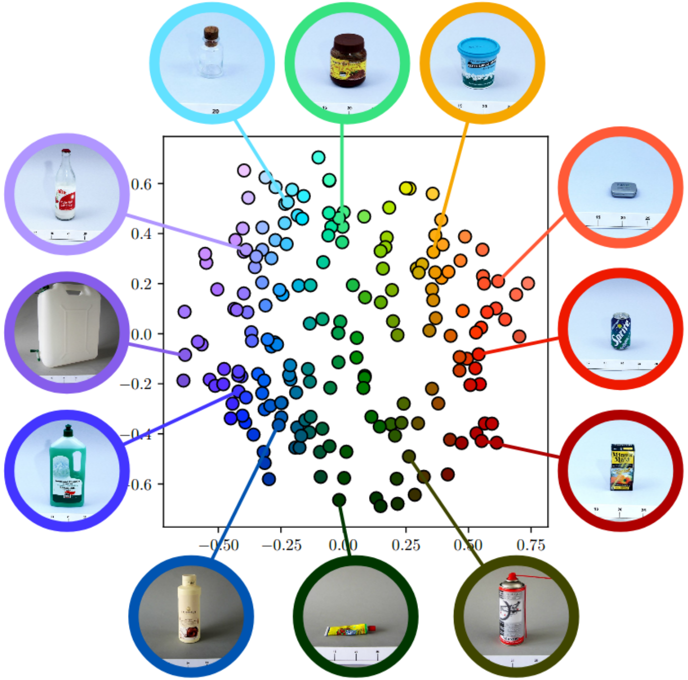
Kemp & Regier 2012, *Science*

Spatial terms



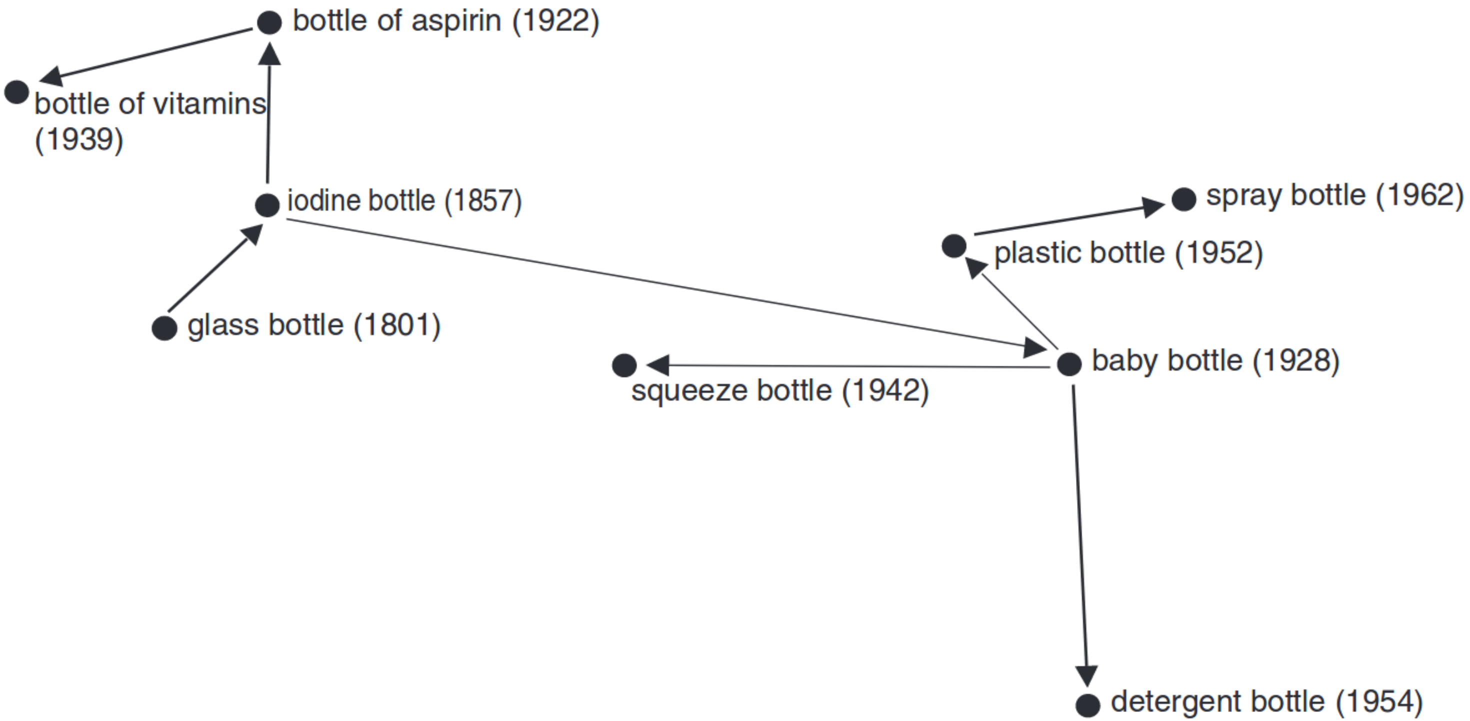
Khetarpal et al. 2013, *Cognitive Science Conference*.

Household containers



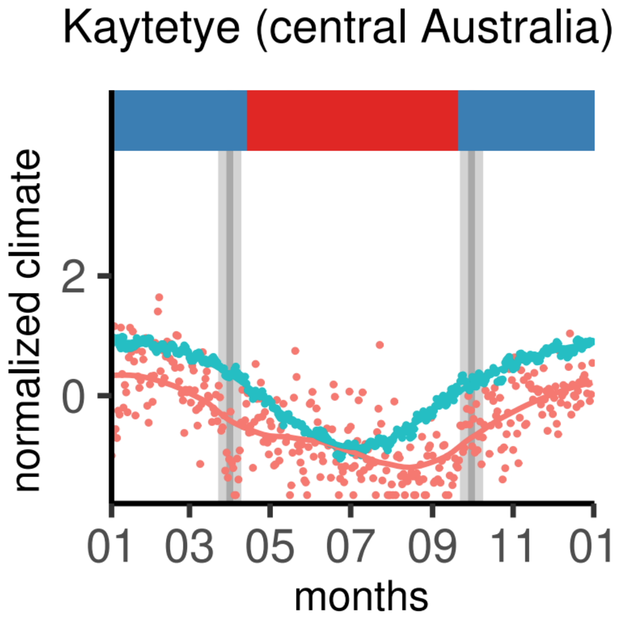
Zaslavsky et al. 2019, *Cognitive Science conference*.

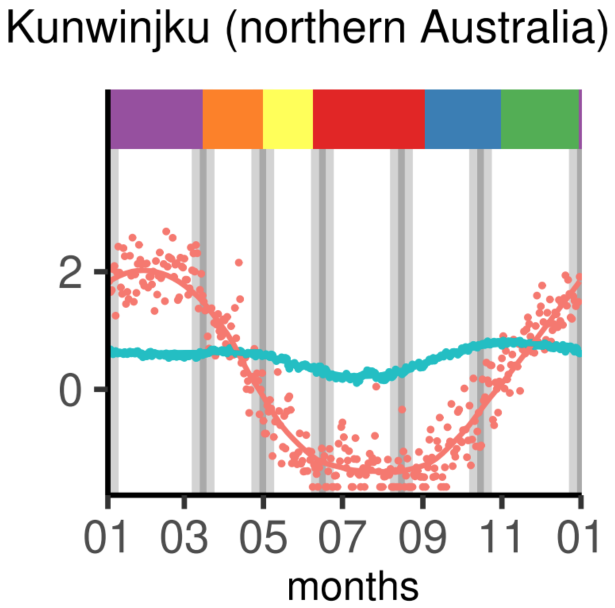
Household containers over time



Xu et al. 2016, *Cognitive Science*.

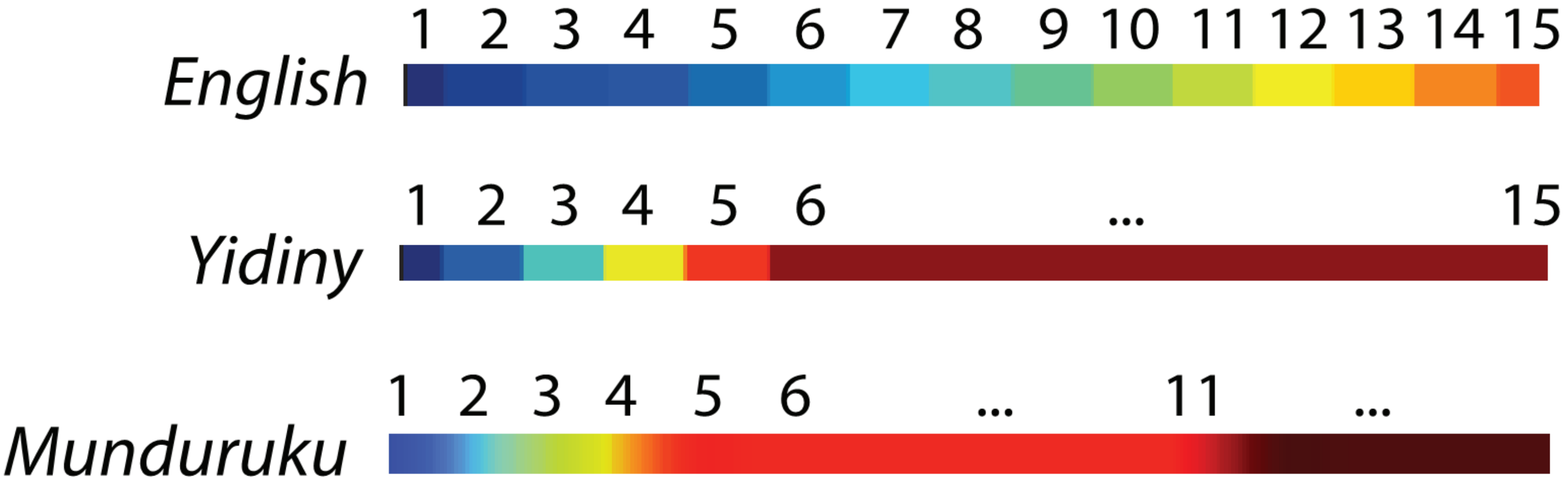
Names for seasons





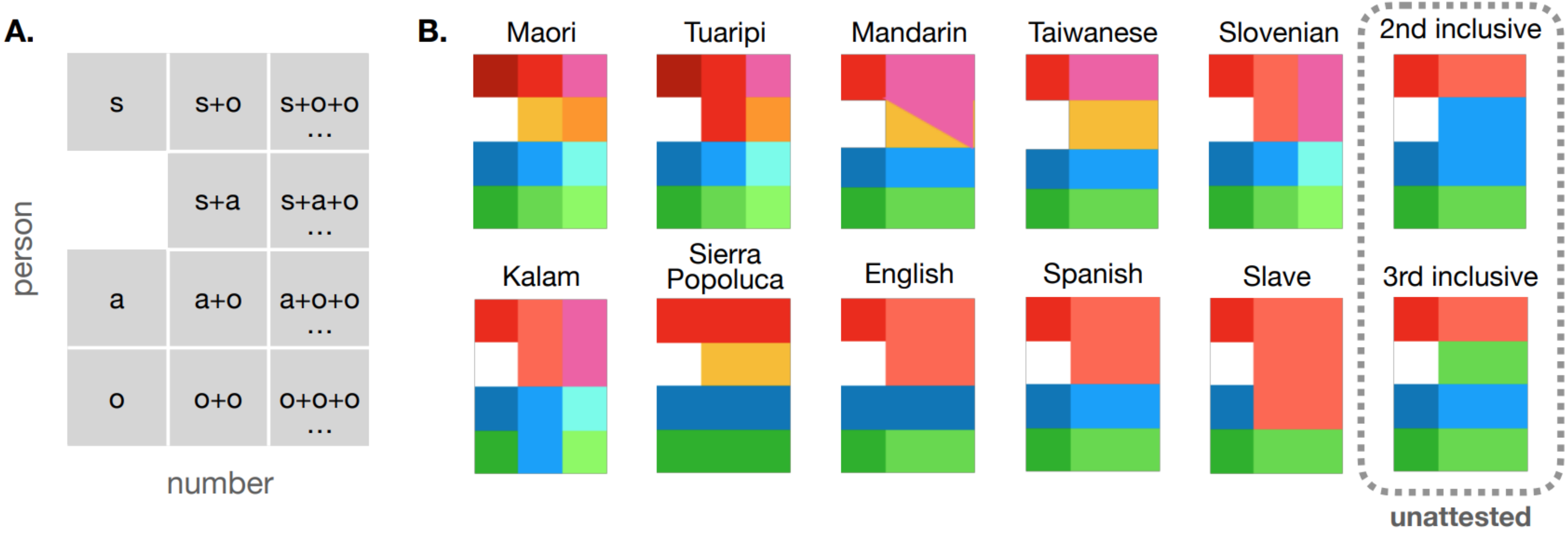
Kemp et al. 2019, *Cognitive Science Conference*.

Numeral systems



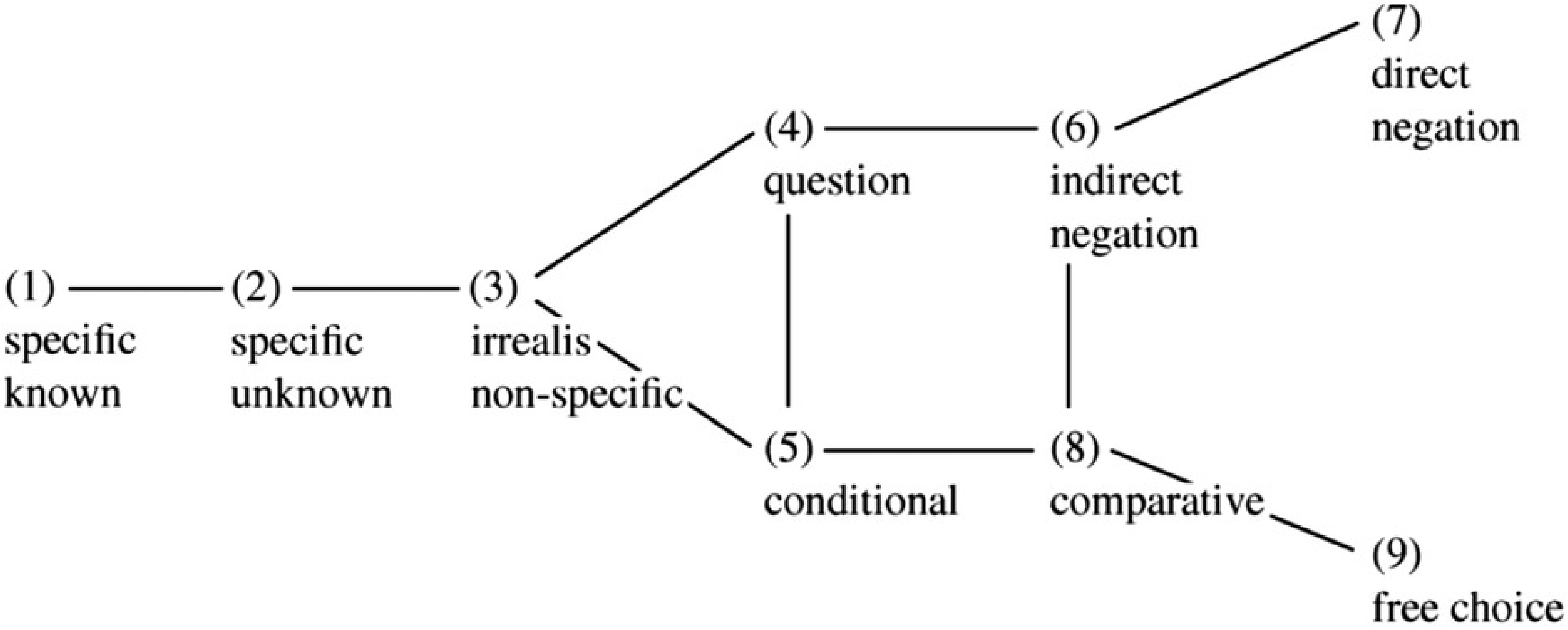
Xu et al. 2020, *Open Mind*.

Personal pronouns



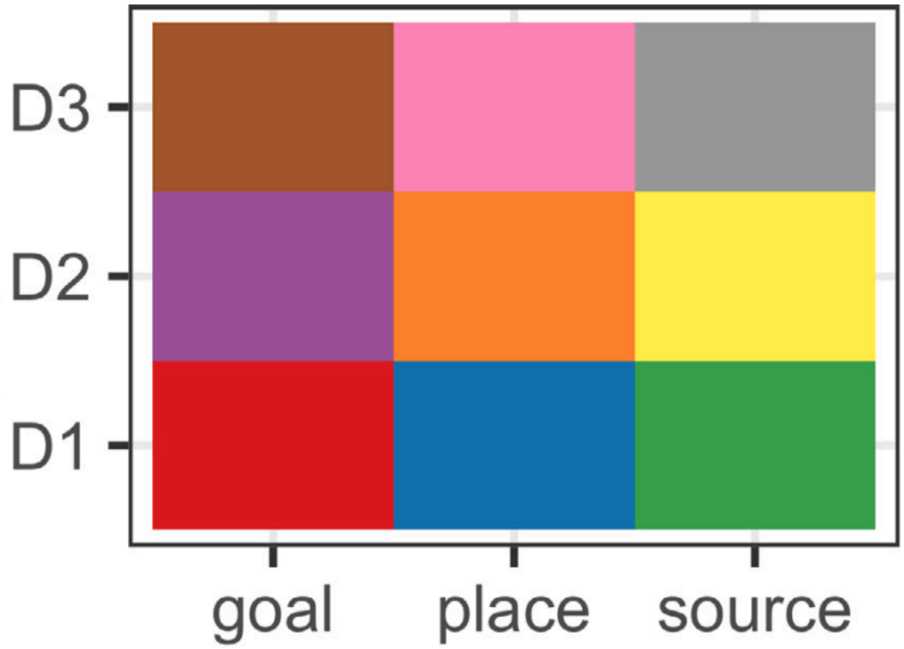
Zaslavsky et al. 2021, *Cognitive Science conference*

Indefinite pronouns



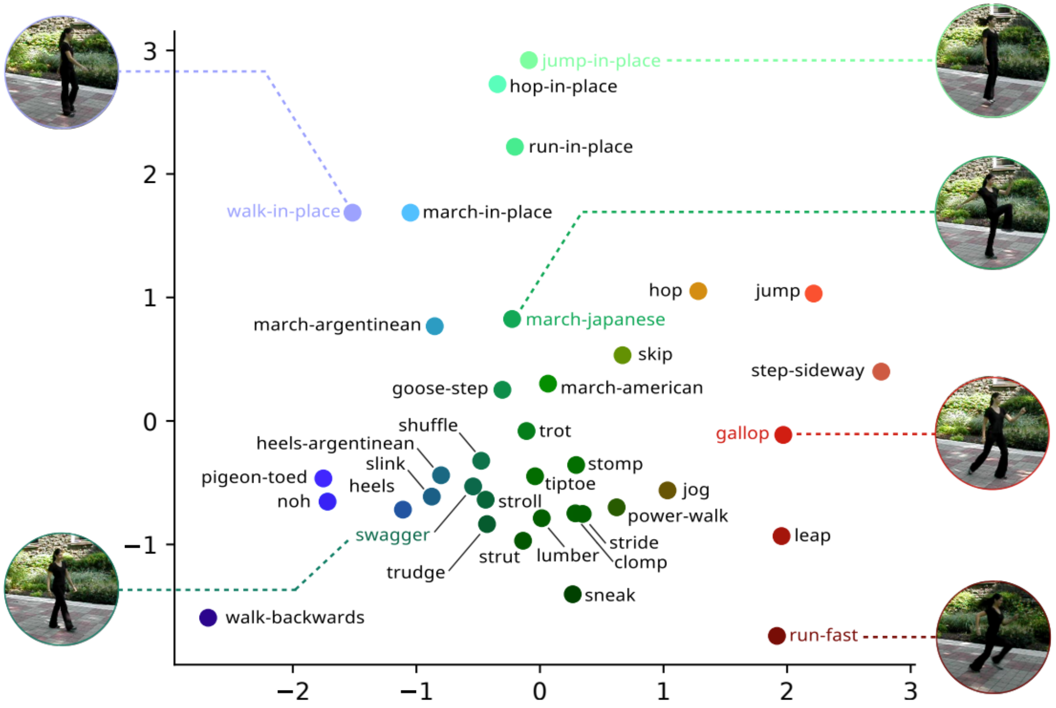
Denić et al. 2022, *Cognitive Science*

Spatial demonstratives



Chen et al. 2023, *Cognition*

Locomotion verbs



Langlois et al. 2025, *Cognitive Science conference*

- Semantic systems across languages may reflect **functional adaptation** in support of **efficient communication**.

What **questions** do you have?

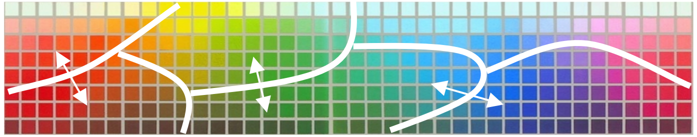
What **comments** do you have?

Wait!

How do **universal focal colors** fit into all of this?

Relativist

No universal foci: categories organized at boundaries by linguistic convention. (Roberson, Davies, & Davidoff, 2000; Roberson, Davidoff, Davies, & Shapiro, 2005).



Boundaries free to vary, and memory varies with them.

Which is itself constrained by the shape of color space and the need to communicate efficiently.

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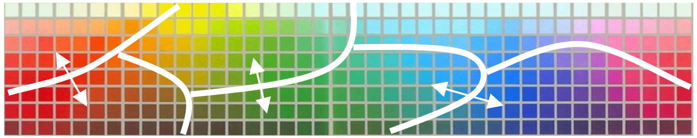

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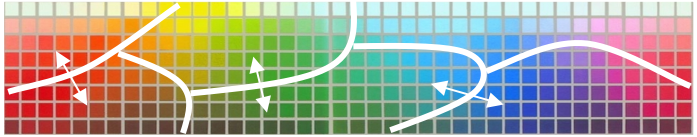
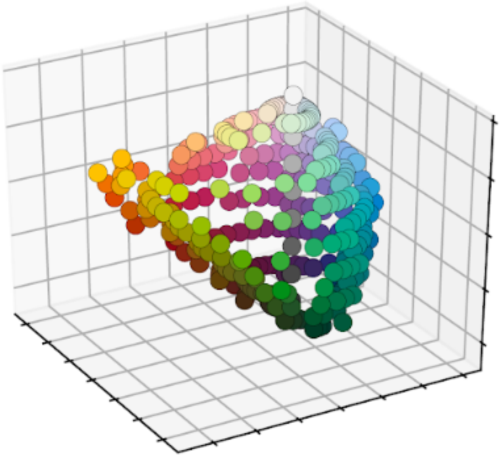


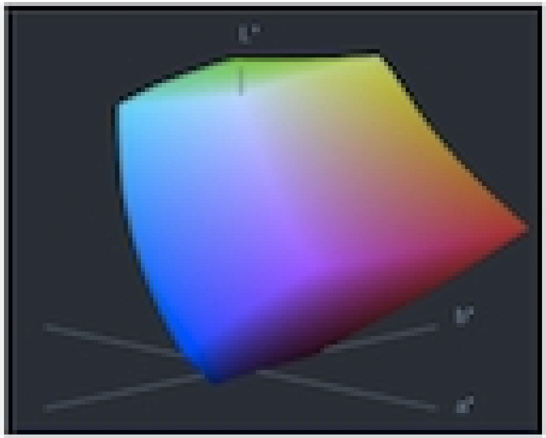
not as free as you might think, but some wiggle room

Boundaries free to vary, and memory varies with them.



The perceptual color solid is irregularly shaped

(Jameson and D’Andrade 1997)



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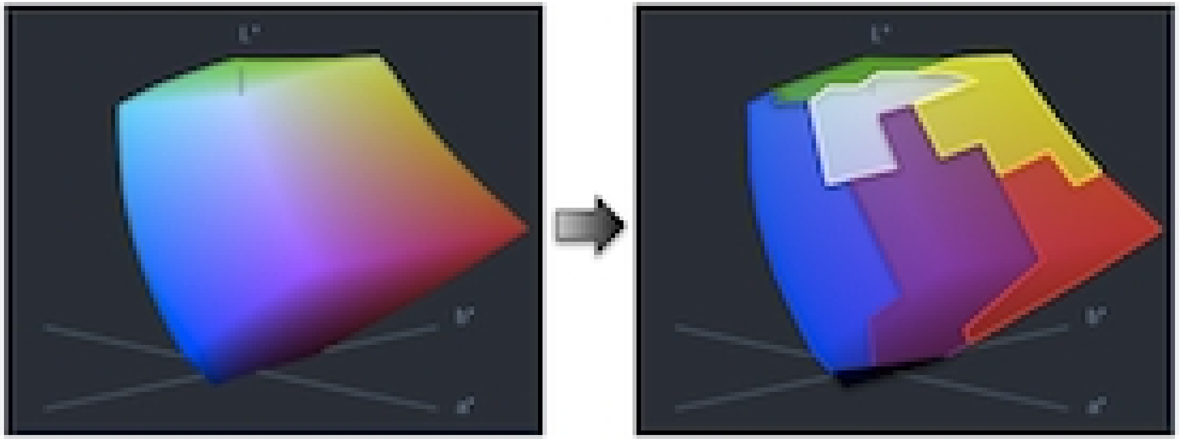
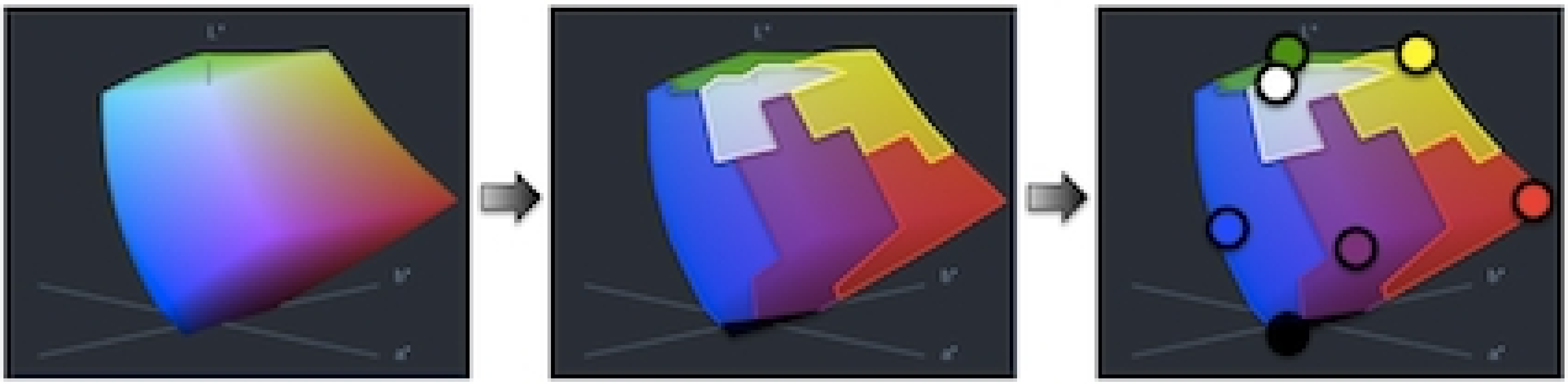


Languages divide it up in various ways, but always under functional pressure to yield categories that are useful: informative and **efficient** (Jameson and D'Andrade 1997; Zaslavsky et al., 2018)

Once those categories have been determined, their best examples can be derived from them as maximally **representative** members of the category (Abbott et al. 2016, *PNAS*)



Thanks to Josh Abbott for these images, and the research behind them.

joshua abbott

joshua.t.abbott@gmail.com

Adobe

- The universalist-relativist framing that we started with is a useful starting point – but it is **reductive** and **simplistic**.

- The universalist-relativist framing that we started with is a useful starting point – but it is **reductive** and **simplistic**.
- Something more **interestingly complicated** seems to be at play.

What **questions** do you have?

What **comments** do you have?